CollectX - Collections & Recovery Management System

1. Introduction

CollectX is a web-based **Collections & Recovery Management System** designed for banks, NBFCs, and fintech lenders to manage end-to-end delinquency handling—from **strategy segmentation, queue allocation, dunning orchestration, promise-to-pay (PTP) tracking, field collections, payment posting references, settlements, repossessions, to write-off and post write-off recoveries**.

It supports **retail and SME loan products** (secured/unsecured) and integrates with core loan servicing (Phase-1 as internal references only). The platform provides **risk-based strategies**, **contact policies**, **agent performance tracking**, and **regulatory grade audit trails**.

It uses a **REST API-based backend** and **Angular/React** frontend, compatible with **Java (Spring Boot)** or **.NET (ASP.NET Core)**.

Actors / Users

- **Collections Agent (Inbound/Outbound)**: Works assigned queues, captures contacts, PTPs, and payments.
- **Field Collections Officer**: Executes field visits, repossessions, and document serving.
- **Collections Supervisor**: Manages strategy assignment, queue balancing, and approvals.
- **Recovery Manager**: Manages settlements, legal/agency placements, post write-off recoveries.
- **Compliance & QA Analyst**: Reviews call logs, contact policy adherence, audit trails.
- **Admin**: Configures products, strategies, scorecards, users/roles, and reports.

2. Module Overview

2.1 Identity & Access Management
Authentication, RBAC, and audit logs.

2.2 Portfolio Ingestion & Delinquency Staging
Ingest loan accounts, compute DPD, bucketization, risk score references.

2.3 Strategy, Segmentation & Queue Allocation
Strategy rules, segmentation, agent capacity and territory allocation.

2.4 Dunning Orchestration & Contact Policies
Channel cadence (in-app placeholders), contact attempts, consent and preferences.

2.5 Agent Workbench & Case Handling
360-view, call logs, PTPs, disputes, hardship flags, follow-ups.

2.6 Payments, PTP & Settlements
Promise management, payment references, settlements/waivers approvals.

2.7 Field Collections, Repossession & Agency Management
Visit scheduling, GPS notes (ref), repossession workflow, agency placements.

2.8 Legal, Write-Off & Post Write-Off Recoveries
Notices, legal case references, charge-off, recovery accounting references.

2.9 Reporting, KPIs & Performance Management
Roll rates, cure rates, PTP kept, cash collected, agent productivity.

2.10 Notifications & Alerts
PTP due, bucket roll risk, approvals, compliance flags.

3. Architecture Overview

- **Frontend**: Angular or React
- **Backend**: REST API microservices
- **Database**: Relational (MySQL/PostgreSQL/SQL Server)
- **Integrations (Phase-1)**: Core loan & payment systems **simulated as references**; dialer/SMS/email **not integrated** (in-app only)

4. Module-Wise Design

4.1 Identity & Access Management Module

Features:

- User login, RBAC, session management, audit trails

Entities:

- **User**
  - UserID
  - Name
  - Role (Agent/Supervisor/Field/Recovery/Compliance/Admin)
  - Email
  - Phone
  - Status (Active/Inactive/Locked)
- **AuditLog**
  - AuditID

    - UserID
    - Action
    - Resource
    - Timestamp
    - Metadata

4.2 Portfolio Ingestion & Delinquency Staging Module

Features:

- Import delinquent loans, compute DPD/buckets, stage accounts

Entities:

- **LoanRef**
    - LoanAccountID
    - CustomerID
    - Product
    - PrincipalOS
    - InterestOS
    - Bucket (0-30/31-60/61-90/90+)
    - DPD
    - LastPaymentDate
    - Region
    - Status (Current/Delinquent/NPA)
- **RiskRef**
    - RiskID
    - LoanAccountID
    - BureauScore
    - InternalScore
    - RiskBand (Low/Med/High)
    - AsOfDate
- **DelinquencyStage**
    - StageID
    - LoanAccountID

    - Bucket
    - StageDate
    - ReasonCode (BrokenPTP/NSF/HARDSHIP/OTHER)

4.3 Strategy, Segmentation & Queue Allocation Module

Features:

- Rule-driven segmentation; assign accounts to queues/agents

Entities:

- **StrategyRule**
    - RuleID
    - Name
    - Expression (JSON/DSL)
    - Priority
    - Status (Active/Inactive)
- **Queue**
    - QueueID
    - Name
    - BucketScope
    - Territory
    - Capacity
    - Status
- **Assignment**
    - AssignmentID
    - LoanAccountID
    - QueueID
    - AgentID
    - AssignedDate
    - Status (Open/Reassigned/Closed)

4.4 Dunning Orchestration & Contact Policies Module

Features:

- Define cadence by bucket; contact attempts with consent flags (in-app log only)

Entities:

- **ContactPolicy**
  - PolicyID
  - Bucket
  - MaxAttemptsPerDay
  - MinGapMinutes
  - PreferredChannelsJSON
  - DoNotCallFlag (Yes/No)
- **ContactAttempt**
  - AttemptID
  - LoanAccountID
  - AgentID
  - Channel (Call/Visit/InApp)
  - AttemptTime
  - Outcome (Connected/LeftMsg/NoAnswer/Refused)
  - Notes
- **ConsentPref**
  - ConsentID
  - CustomerID
  - Channel (Call/SMS/Email/App)
  - Status (Allowed/OptOut)
  - UpdatedDate

4.5 Agent Workbench & Case Handling Module

Features:

- Single-pane 360; disputes/hardship; follow-ups and tasks

Entities:

- **CollectionCase**
  - CaseID

    - LoanAccountID
    - OpenedDate
    - CaseType (Delinquency/Dispute/Hardship)
    - Priority (Low/Med/High)
    - Status (Open/InProgress/Resolved/Closed)
- **CaseNote**
    - NoteID
    - CaseID
    - AuthorID
    - NoteText
    - CreatedDate
- **FollowUpTask**
    - TaskID
    - CaseID
    - DueDate
    - TaskType (Call/Visit/Docs/PTPCheck)
    - AssignedTo
    - Status (Open/Done/Overdue)
- **HardshipFlag**
    - HardshipID
    - LoanAccountID
    - Reason (Medical/JobLoss/NaturalCalamity)
    - StartDate
    - EndDate
    - Status

4.6 Payments, PTP & Settlements Module

Features:

- Capture PTPs, monitor kept/broken; payment references; settlements & waivers

Entities:

- **PTP**
    - PTPID
    - LoanAccountID
    - PromisedAmount
    - PromiseDate
    - PromisedBy
    - Status (Open/Kept/Broken)
- **PaymentRef**
    - PaymentRefID
    - LoanAccountID
    - Amount
    - PaidDate
    - Mode (Transfer/UPI/Cash/Cheque)
    - ReferenceNo
    - Status (Posted/Failed)
- **Settlement**
    - SettlementID
    - LoanAccountID
    - OfferAmount
    - WaiverAmount
    - TermsJSON
    - ApprovalStatus (Requested/Approved/Rejected)
    - ApprovalBy
    - ApprovalDate
    - Status (Active/Expired/Honored/Defaulted)

4.7 Field Collections, Repossession & Agency Management Module

Features:

- Visit scheduling, repossession steps, external agency placement

Entities:

- **FieldVisit**

    - VisitID
    - LoanAccountID
    - AgentID
    - ScheduledDate
    - Outcome (Met/NotMet/AddressChanged)
    - Notes
    - GeoTag (Ref)
- **Repossession** *(secured loans)*
    - RepossessionID
    - LoanAccountID
    - AssetDetailsJSON
    - InitiatedDate
    - ActionStatus (Initiated/Seized/Stored/Auctioned/Released)
    - RealizationAmount
- **Agency**
    - AgencyID
    - Name
    - Territory
    - CommissionPct
    - Status (Empaneled/Suspended)
- **AgencyPlacement**
    - PlacementID
    - LoanAccountID
    - AgencyID
    - StartDate
    - EndDate
    - CommissionBasis (Collected/Slab)
    - Status (Active/Closed)

4.8 Legal, Write-Off & Post Write-Off Recoveries Module

Features:

- Legal notices, suits (reference), charge-off process, recoveries

Entities:

- **LegalAction**
  - LegalID
  - LoanAccountID
  - ActionType (Notice/Arbitration/Suit)
  - FiledDate
  - Status (Open/Disposed)
  - Notes
- **WriteOff**
  - WriteOffID
  - LoanAccountID
  - ChargeOffDate
  - PrincipalWO
  - InterestWO
  - FeesWO
  - Status (Posted/Reversed)
- **Recovery**
  - RecoveryID
  - LoanAccountID
  - Amount
  - RecoveryDate
  - Source (Agency/Legal/WalkIn)
  - Notes

4.9 Reporting, KPIs & Performance Management Module

Features:

- Cash collected, PTP kept %, cure rate, roll rate, agency yield, field productivity

Entities:

- **CollectionsReport**

    - ReportID
    - Scope (Bucket/Territory/Product/Period)
    - Metrics (CashCollected, PTPKeptPct, CureRate, RollRate, Realization)
    - GeneratedDate
- **AgentPerformance**
    - PerfID
    - AgentID
    - Period
    - AccountsWorked
    - ContactsMade
    - PTPsBooked
    - PTPKept
    - AmountCollected

4.10 Notifications & Alerts Module

Features:

- Alerts for PTP due, broken PTP, approvals, roll risk, compliance

Entities:

- **Notification**
    - NotificationID
    - UserID
    - Message
    - Category (PTP/Approval/Compliance/Queue)
    - Status (Unread/Read/Dismissed)
    - CreatedDate

5. Deployment Strategy

- **Local**: Angular/React frontend + Spring Boot/.NET Core backend + local DB
- **Production**: Cloud/on-prem, secure APIs, stateless services, job schedulers for staging, strategy runs, and nightly KPI packs

6. Database Design

Tables:

- User(UserID, Name, Role, Email, Phone, Status)
- AuditLog(AuditID, UserID, Action, Resource, Timestamp, Metadata)
- LoanRef(LoanAccountID, CustomerID, Product, PrincipalOS, InterestOS, Bucket, DPD, LastPaymentDate, Region, Status)
- RiskRef(RiskID, LoanAccountID, BureauScore, InternalScore, RiskBand, AsOfDate)
- DelinquencyStage(StageID, LoanAccountID, Bucket, StageDate, ReasonCode)
- StrategyRule(RuleID, Name, Expression, Priority, Status)
- Queue(QueueID, Name, BucketScope, Territory, Capacity, Status)
- Assignment(AssignmentID, LoanAccountID, QueueID, AgentID, AssignedDate, Status)
- ContactPolicy(PolicyID, Bucket, MaxAttemptsPerDay, MinGapMinutes, PreferredChannelsJSON, DoNotCallFlag)
- ContactAttempt(AttemptID, LoanAccountID, AgentID, Channel, AttemptTime, Outcome, Notes)
- ConsentPref(ConsentID, CustomerID, Channel, Status, UpdatedDate)
- CollectionCase(CaseID, LoanAccountID, OpenedDate, CaseType, Priority, Status)
- CaseNote(NoteID, CaseID, AuthorID, NoteText, CreatedDate)
- FollowUpTask(TaskID, CaseID, DueDate, TaskType, AssignedTo, Status)
- HardshipFlag(HardshipID, LoanAccountID, Reason, StartDate, EndDate, Status)
- PTP(PTPID, LoanAccountID, PromisedAmount, PromiseDate, PromisedBy, Status)
- PaymentRef(PaymentRefID, LoanAccountID, Amount, PaidDate, Mode, ReferenceNo, Status)
- Settlement(SettlementID, LoanAccountID, OfferAmount, WaiverAmount, TermsJSON, ApprovalStatus, ApprovalBy, ApprovalDate, Status)
- FieldVisit(VisitID, LoanAccountID, AgentID, ScheduledDate, Outcome, Notes, GeoTag)
- Repossession(RepossessionID, LoanAccountID, AssetDetailsJSON, InitiatedDate, ActionStatus, RealizationAmount)

- Agency(AgencyID, Name, Territory, CommissionPct, Status)
- AgencyPlacement(PlacementID, LoanAccountID, AgencyID, StartDate, EndDate, CommissionBasis, Status)
- LegalAction(LegalID, LoanAccountID, ActionType, FiledDate, Status, Notes)
- WriteOff(WriteOffID, LoanAccountID, ChargeOffDate, PrincipalWO, InterestWO, FeesWO, Status)
- Recovery(RecoveryID, LoanAccountID, Amount, RecoveryDate, Source, Notes)
- CollectionsReport(ReportID, Scope, Metrics, GeneratedDate)
- AgentPerformance(PerfID, AgentID, Period, AccountsWorked, ContactsMade, PTPsBooked, PTPKept, AmountCollected)
- Notification(NotificationID, UserID, Message, Category, Status, CreatedDate)

7. User Interface Design

- **Agent Workbench**: Assigned accounts, 360-view, contact attempts, PTP, tasks.
- **Supervisor Console**: Strategy runs, queue capacity, reassignments, approvals.
- **Field App (Web/Mobile)**: Today's visits, outcome capture, geo-tag (ref), repossession steps.
- **Recovery Desk**: Settlements, waivers, legal, agency placements, write-offs.
- **Compliance/QA**: Call attempt logs, contact policy adherence, hardship oversight.
- **Reports & KPIs**: Roll/cure, PTP kept, cash collected, agent productivity, aging.
- **Admin Panel**: Products, strategy rules, territories, users/roles, reports.

8. Non-Functional Requirements

- **Performance**: Handle **50,000 concurrent users**; fast queue fetch and PTP actions.
- **Security**: PII encryption, RBAC, audit logs, least-privilege access.
- **Scalability**: Portfolio sizes in millions; nightly strategy/staging batches.
- **Availability**: **99.9% uptime**; degraded mode for reporting during batch runs.
- **Maintainability**: Modular services, versioned APIs, configuration-driven strategies.
- **Observability**: Centralized logs; KPIs (PTP kept %, cure rate, roll rate, agent productivity); alerting.

9. Assumptions & Constraints

- Phase-1 excludes real dialer/SMS/email integrations—**in-app attempts only**.
- No direct core posting in Phase-1; **PaymentRef** stores reference data for reconciliation.
- Geo-tagging and device telemetry are stored as **references only** (no SDK integration).
- Legal case management is **basic** (external court systems not integrated).
- Fully implementable using **Java Spring Boot/ASP.NET Core + Angular/React** with a relational DB.